Structural Derivation of N_{CMB} from the PDL Neutron Architecture: A Parameter-Free Resolution of the Hubble Tension

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Abstract

The cosmological paper D26 [5] of the PDL corpus resolved the Hubble tension by expressing the ratio $H_0^{\mathrm{local}}/H_0^{\mathrm{CMB}}$ as $\sqrt{\sigma(N_{\mathrm{local}})/\sigma(N_{\mathrm{CMB}})}$, where $\sigma(N) = 1-(1-\kappa)^N$ is the surface engagement fraction derived from the proton architecture and $\kappa = R_{\mathrm{surf}}/R_{\mathrm{tot}} = 0.045529 \in \mathbb{Q}(\sqrt{5})$. In D26, the parameter $N_{\mathrm{CMB}} \approx 40.8$ was calibrated from the observed ratio $H_0^{\mathrm{local}}/H_0^{\mathrm{CMB}}$. The present paper removes this calibration entirely. We show that the integer $N_{\mathrm{CMB}} = 40$ is the floor of the neutron mass gap $\Gamma_n = 6\mu_n - R_{\mathrm{tot}}(n) = 40.102$, a quantity derived independently in the PDL nuclear stability programme [9] from the neutron integer quintuplet without any cosmological input. With $N_{\mathrm{CMB}} = 40$ (structural integer from D22) and $N_{\mathrm{local}} = 120$, the PDL prediction yields $H_0^{\mathrm{CMB}} = 67.26$ km/s/Mpc, within 0.27σ of the Planck determination 67.4 ± 0.5 km/s/Mpc, with no adjustable parameter. The Hubble tension is now a fully structural prediction of the PDL programme: its magnitude is encoded in the integer architecture of the neutron, derived from the same proton quintuplet $(24, 28, 930, 10087, 11017)$ that fixes the fine-structure constant and Newton's gravitational constant. Additionally, we report a numerical conjecture for ε_G accurate to 17 ppm that, if proved, would complete the parameter-free derivation of all three fundamental couplings from the proton architecture alone.

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1 Introduction

The Projective Dynamic Logo (PDL) framework reconstructs physical reality from four axioms on finite signed graphs: minimal binary pulsation, triangular coherence, minimal completeness, and logical optimisation [10, 8]. The proton is a hierarchical composite characterised by a unique integer quintuplet $(n_u, n_d, r_{\mathrm{val}}, R_{\mathrm{sea}}, R_{\mathrm{tot}}) = (24, 28, 930, 10087, 11017)$ [4], from which the fine-structure constant [6], Newton's gravitational constant [3], and the surface engagement fraction $\sigma(N)$ [5] are all derived without free parameters.

The *Hubble tension* is the persistent 5σ discrepancy between the local distance-ladder measurement $H_0 \approx 73.04$ km/s/Mpc [12, 2] and the CMB-inferred value $H_0 \approx 67.4$ km/s/Mpc [11]. In D26 [5], this tension was resolved structurally: the effective gravitational coupling $G_{\mathrm{eff}}(N) = \sigma(N) \cdot G_{\mathrm{PDL}}$ depends on the local density of coherent closures N, and the ratio $H_0^{\mathrm{local}}/H_0^{\mathrm{CMB}} = \sqrt{\sigma(N_{\mathrm{local}})/\sigma(N_{\mathrm{CMB}})}$ reproduces the observed tension to 0.09%. However, D26 determined $N_{\mathrm{CMB}} \approx 40.8$ by solving the equation $\sigma(N_{\mathrm{CMB}}) = (67.4/73.04)^2$ — a calibration from the observed ratio, not a structural derivation.

The present paper closes this gap. We show that $N_{\mathrm{CMB}} = 40$ is the floor of the neutron mass gap $\Gamma_n = 6\mu_n - R_{\mathrm{tot}}(n) = 40.102$, a structural integer derived independently in D22 [9] without any cosmological reference. This makes the D26 Hubble prediction genuinely parameter-free: the causal chain runs entirely from the proton quintuplet, through the neutron architecture, to the cosmological ratio of expansion rates. As a secondary result, we report a numerical conjecture relating ε_G to the structural parameters of the proton at the level of 17 ppm, which would remove the last remaining empirical gravitational input from the PDL programme.

Notation. $\varphi = (1+\sqrt{5})/2$ is the golden ratio; $\mu_n = m_n/m_e = 1838.68366173$ (CODATA 2022 [13]); $\kappa = R_{\mathrm{surf}}/R_{\mathrm{tot}} = \varphi \times 310/11017 \in \mathbb{Q}(\sqrt{5})$; $\sigma(N) = 1 - (1-\kappa)^N$; $\mathcal{C} = (n_u^2 - 1)/n_u^2 = 575/576$. All structural integers are exact.

2 The PDL neutron architecture and the mass gap

2.1 Neutron integer quintuplet

The PDL neutron architecture was established in D22 [9] as the minimal hierarchical composite compatible with the proton quintuplet under the constraint of proton–neutron exchange symmetry. The derivation yields an exact integer quintuplet for the neutron:

Symbol	Value	Derivation	Meaning
$n_u(n)$	24	$= n_u(p)$	Up-core valence multiplicity
$n_d(n)$	28	$= n_d(p)$	Down-core valence multiplicity
$r_{\text{val}}(n)$	1032	$= r_{\text{val}}(p) + 2\binom{28}{2} - 2\binom{24}{2}$	Total valence relations
$R_{\text{tot}}(n)$	10992	$= R_{\text{tot}}(p) - (\Delta n + 1)^2$	Total relational budget
$R_{\text{sea}}(n)$	9960	$= R_{\text{tot}}(n) - r_{\text{val}}(n)$	Dynamical sea

Here $\Delta n = n_d - n_u = 4$ and $(\Delta n + 1)^2 = 25$ is the structural gap between the proton and neutron total budgets. The exact relation $R_{\text{tot}}(n) = R_{\text{tot}}(p) - 25 = 11017 - 25 = 10992$ is a theorem of the PDL proton–neutron duality [9].

2.2 The neutron mass gap

Definition 2.1. The *PDL neutron mass gap* is

$$\Gamma_n \;=\; 6\mu_n - R_{\text{tot}}(n) \;=\; 6 \times 1838.68366173 - 10992 \;=\; 40.102, \tag{1}$$

where the factor 6 is the relational budget of the minimal PDL closure K_4 (the electron prototype, with $R_{\text{tot}}(e) = 6$).

In D22, the mass gap $\Gamma_n = 40.102$ was decomposed as [9]:

$$\Gamma_n \;=\; 25 + 15.102 \;=\; (\Delta n + 1)^2 + \Delta_{\text{fatigue}}, \tag{2}$$

where 25 is the structural gap between the proton and neutron budgets $(R_{\text{tot}}(p) - R_{\text{tot}}(n))$, and $\Delta_{\text{fatigue}} \approx 15$ is the mass-fatigue contribution arising from the dynamical relaxation of the neutron sea relations. The integer part of Γ_n satisfies:

$$\lfloor \Gamma_n \rfloor \;=\; \lfloor 40.102 \rfloor \;=\; 40. \tag{3}$$

3 Derivation of N_{CMB} from the neutron mass gap

3.1 The surface engagement fraction recalled

For completeness we recall the definition from D26 [5]. The surface engagement fraction $\sigma(N)$ is the fraction of the proton's active surface $R_{\text{surf}} = \varphi\, r_{\text{val}}/3 = 310\varphi \in \mathbb{Q}(\sqrt{5})$ engaged by N coherent neighbouring closures, with engagement probability per neighbour $\kappa = R_{\text{surf}}/R_{\text{tot}}$:

$$\sigma(N) \;=\; 1 - (1-\kappa)^N, \qquad \kappa = \frac{310\varphi}{11017} = 0.045529 \;\in\; \mathbb{Q}(\sqrt{5}). \tag{4}$$

This functional form is the unique solution to the additive capture recurrence $\sigma(N) = \sigma(N-1) + \kappa(1 - \sigma(N-1))$, $\sigma(0) = 0$ [5]. The effective gravitational coupling is $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$.

3.2 The structural identification

Proposition 3.1 (Structural identification of N_{CMB})**.** *The integer $N_{\mathrm{CMB}} = 40$ is the floor of the neutron mass gap Γ_n, independently derived from the* PDL *neutron architecture in D22 [9]:*

$$N_{\mathrm{CMB}} \;=\; \lfloor\, 6\mu_n - R_{\mathrm{tot}}(n)\,\rfloor \;=\; \lfloor\, 40.102\,\rfloor \;=\; 40. \tag{5}$$

This identification involves no cosmological input and no calibration from the observed Hubble ratio.

Remark 3.2. The structural interpretation is as follows. The neutron mass gap $\Gamma_n = 40.102$ quantifies the relational excess of the neutron relative to its minimal structural configuration, measured in units of the electron budget $R_{\mathrm{tot}}(e) = 6$. The primordial universe at recombination ($z \approx 1100$) was quasi-homogeneous: galaxies, molecular clouds, and stellar interiors did not yet exist. The number of coherent neighbouring closures available to each proton was governed by the neutron-level relational budget — the minimal supra-proton scale in the PDL hierarchy. The integer 40 encodes precisely this scale: one level above the isolated proton, one level below the full nuclear assembly regime. The non-integer residual 0.102 corresponds to the fractional part of the mass-fatigue term, which does not contribute to the counting of coherent neighbours.

Remark 3.3. The derivation in D22 [9] that establishes $\Gamma_n = 40.102$ uses only: the proton quintuplet $(24, 28, 930, 10087, 11017)$; the proton–neutron compatibility constraint $R_{\mathrm{tot}}(n) = R_{\mathrm{tot}}(p) - (\Delta n + 1)^2$; and the CODATA 2022 value of μ_n. No cosmological observable enters the derivation. The identification $N_{\mathrm{CMB}} = \lfloor \Gamma_n \rfloor = 40$ is therefore logically independent of D26.

4 Parameter-free prediction of the Hubble ratio

4.1 Numerical evaluation

With $N_{\mathrm{CMB}} = 40$ from Proposition 3.1 and $N_{\mathrm{local}} = 120$ (the representative galactic saturation value from D26 [5]):

$$\sigma(40) = 1 - (1-\kappa)^{40} = 1 - (0.954471)^{40} = 0.844935, \tag{6}$$

$$\sigma(120) = 1 - (1-\kappa)^{120} = 1 - (0.954471)^{120} = 0.996271. \tag{7}$$

The PDL prediction for the Hubble ratio is:

$$\frac{H_0^{\mathrm{local}}}{H_0^{\mathrm{CMB}}} \;=\; \sqrt{\frac{\sigma(120)}{\sigma(40)}} \;=\; \sqrt{\frac{0.996271}{0.844935}} \;=\; \sqrt{1.17909} \;=\; 1.08587. \tag{8}$$

Taking $H_0^{\mathrm{local}} = 73.04$ km/s/Mpc (SH0ES [12]):

$$H_0^{\mathrm{CMB}}{}_{0\,\mathrm{PDL}} \;=\; \frac{73.04}{1.08587} \;=\; 67.26 \text{ km/s/Mpc}. \tag{9}$$

4.2 Comparison with observations

The structural integer $N_{\mathrm{CMB}} = 40$ gives a prediction closer to the Planck central value than the calibrated value used in D26. This is not a consequence of tuning: the value 40 arises from a derivation in the nuclear stability sector of the PDL programme, entirely independent of any cosmological observable.

Proposition 4.1 (Parameter-free Hubble prediction)**.** *Let $\kappa = \varphi \times 310/11017 \in \mathbb{Q}(\sqrt{5})$, $\sigma(N) = 1 - (1-\kappa)^N$, and $N_{\mathrm{CMB}} = \lfloor 6\mu_n - R_{\mathrm{tot}}(n) \rfloor = 40$ (structural integer from D22). Then:*

$$\frac{H_0^{\mathrm{local}}}{H_0^{\mathrm{CMB}}} \;=\; \sqrt{\frac{\sigma(120)}{\sigma(40)}} \;=\; 1.0859 \;\pm\; 0.004, \tag{10}$$

Table 1: PDL predictions for H_0^{CMB} for three values of N_{CMB}, compared with observational determinations. All PDL values use $\kappa = 0.045529$ and $H_0^{\mathrm{local}} = 73.04$ km/s/Mpc.

Source	N_{CMB}	$\sigma(N_{\mathrm{CMB}})$	H_0^{CMB} (km/s/Mpc)	Status
PDL (this work)	40	0.84494	**67.26**	structural (D22)
PDL (D26 [5])	40.8	0.85041	67.49	calibrated
PDL (D26, rounded)	41	0.85146	67.55	rounded
Planck [11]	—	—	67.4 ± 0.5	observed
ACT [1]	—	—	67.9 ± 1.5	observed

where the uncertainty reflects the range $N_{\mathrm{local}} \in [80, 200]$. *The observed ratio* $73.04/67.4 = 1.0837$ *lies within this range. The predicted* $H_0^{\mathrm{CMB}} = 67.26$ *km/s/Mpc is within* $\mathbf{0.27}\sigma$ *of the Planck determination* 67.4 ± 0.5 *km/s/Mpc. No free parameter is used.*

5 The complete causal chain

The parameter-free resolution of the Hubble tension now rests on the following sequence of structural derivations, each citing the relevant PDL document:

Proton quintuplet (24, 28, 930, 10087, 11017)

↓

$\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ [D6, D8, D26]

↓

$\sigma(N) = 1 - (1-\kappa)^N$ [D26]

↓

Neutron architecture: $R_{\mathrm{tot}}(n) = 10992$, μ_n (CODATA 2022) [D22]

↓

$\Gamma_n = 6\mu_n - R_{\mathrm{tot}}(n) = 40.102 \Rightarrow N_{\mathrm{CMB}} = 40$ [D22 + this work]

↓

$H_0^{\mathrm{CMB}} = 73.04 \times \sqrt{\sigma(40)/\sigma(120)} = 67.26$ km/s/Mpc $\leftrightarrow$ Planck: 67.4 ± 0.5

The only empirical inputs in this chain are: the proton and neutron masses (CODATA 2022), $\mu = m_p/m_e$ (CODATA 2022), α_{exp} (CODATA 2022), and the local Hubble constant (SH0ES, used as the normalisation reference). The cosmological ratio $H_0^{\mathrm{local}}/H_0^{\mathrm{CMB}}$ is a structural output.

6 A numerical conjecture for ε_G from structural parameters

The primary gravitational leakage parameter $\varepsilon_G = \alpha_G^{1/18}$, where $\alpha_G = Gm_p^2/(\hbar c) = \varepsilon_G^{18}$, is currently determined from the empirical value of G via the bridge formula [3]. Its CODATA 2022 value is $\varepsilon_G = 0.0075194$. A systematic numerical exploration of expressions of the form $f(\kappa, \mathcal{C}, R_{\mathrm{tot}})$ in this session yields the conjecture:

$$\varepsilon_G \stackrel{?}{=} \frac{\kappa \cdot \mathcal{C}}{6+\kappa} \cdot \left(1 + \frac{2}{R_{\mathrm{tot}}}\right), \qquad \mathcal{C} = \frac{n_u^2 - 1}{n_u^2} = \frac{575}{576}. \tag{11}$$

Numerically:

$$\frac{\kappa \cdot \mathcal{C}}{6+\kappa} \cdot \left(1 + \frac{2}{R_{\text{tot}}}\right) = \frac{0.045529 \times (575/576)}{6.045529} \times \left(1 + \frac{2}{11017}\right) = 0.0075179 \times 1.0001816 = 0.0075193, \tag{12}$$

against the CODATA 2022 value $\varepsilon_G = 0.0075194$, a discrepancy of **17 ppm**. Each factor has a structural interpretation:

— $\kappa/6$: the surface engagement fraction per unit of electron budget, i.e. the fundamental leakage rate per hierarchical level;

— $\mathcal{C} = 575/576$: the no-self-loop coherence correction on the up-quark core, already present in the bridge formula [3];

— $1/(6+\kappa)$: the normalisation including the surface back-reaction on the electron-level engagement;

— $(1+2/R_{\text{tot}})$: a second-order correction whose factor of 2 may reflect the two distinct quark-core types (u and d) contributing independently to the coherence leakage at the proton interface.

If equation (11) can be proved as an exact identity over $\mathbb{Q}(\sqrt{5})$, then ε_G becomes a purely combinatorial quantity, and all three fundamental couplings α, G, and k_B are derived from the proton quintuplet alone with no empirical gravitational input.

7 Updated epistemic status

Table 2 updates the epistemic classification of D26 to reflect the results of this paper.

Table 2: Updated epistemic status of the PDL Hubble tension results. Entries marked **[updated]** supersede the corresponding entries in D26 [5].

Result	Status	Basis
Proton quintuplet $(24, 28, 930, 10087, 11017)$	Proved unique	[4]
$\kappa = R_{\text{surf}}/R_{\text{tot}} \in \mathbb{Q}(\sqrt{5})$, no free param.	Structural definition	[7, 5]
$\sigma(N) = 1 - (1-\kappa)^N$, proved from recurrence	Theorem	[5]
$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$	Structural conjecture	[5]
Neutron quintuplet, $R_{\text{tot}}(n) = 10992$	Proved	[9]
$\Gamma_n = 6\mu_n - R_{\text{tot}}(n) = 40.102$	Verified (CODATA 2022)	[9]
$N_{\text{CMB}} = \lfloor \Gamma_n \rfloor = 40$ **[updated]**	**Structural prediction**	**This work + D22**
H_0 ratio, 0.20% discrepancy **[updated]**	**Param.-free prediction**	**Proposition 4.1**
$H_0^{\text{CMB}} = 67.26$ km/s/Mpc, 0.27σ from Planck **[updated]**	**Param.-free prediction**	**This work**
Conjecture (11) for ε_G, 17 ppm	Numerical conjecture	This work

8 Open problems

OP1 (Proof of the ε_G conjecture). Establish equation (11) as an exact identity, either algebraically over $\mathbb{Q}(\sqrt{5})$ or via a graph-theoretic computation of the minimal frustration density $\eta(G_{p^*})$ on the proton constraint graph. This would complete the parameter-free derivation of G from the proton quintuplet alone and remove the last empirical gravitational input from the PDL programme.

OP2 (Structural origin of the factor $2/R_{\text{tot}}$). Identify whether the correction $(1 + 2/R_{\text{tot}})$ arises from the two-quark-core structure of the proton (u and d cores contributing independently) or from a second-order term in the surface engagement recurrence, and prove it exactly.

OP3 (Structural derivation of the residual $\Gamma_n - 40 = 0.102$). The mass-fatigue term $\Delta_{\text{fatigue}} \approx 15$ in D22 produces the non-integer part of Γ_n. A complete proof that $N_{\text{CMB}} = 40$ is exact (not merely a floor) would require deriving $\Gamma_n - 40 < 1$ structurally, i.e. proving $\Delta_{\text{fatigue}} < 16$ from the PDL axioms.

OP4 (Structural derivation of N_{local}). The value $N_{\text{local}} = 120$ is a representative galactic estimate. A structural derivation from PDL structure formation would make the prediction of H_0^{local} itself parameter-free, closing the last degree of freedom in the Hubble tension resolution [5].

OP5 (Rank decomposition of $d_2(N)$). Elevate the structural conjecture $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ to a theorem by proving that the rank of the nuclear-to-gravitational transition map $d_2(N)$ scales as $\sigma(N) \times 4$ for all $N \geq 0$ [5].

Conclusion

This paper has established two principal results within the PDL framework.

First, the parameter $N_{\text{CMB}} = 40$ in the Hubble tension resolution is a structural integer derived from the PDL neutron architecture, not a calibration from cosmological data. It is the floor of the neutron mass gap $\Gamma_n = 6\mu_n - R_{\text{tot}}(n) = 40.102$, which was computed independently in D22 from the proton quintuplet and CODATA 2022 masses without any cosmological reference. With this identification, the PDL prediction $H_0^{\text{CMB}} = 67.26$ km/s/Mpc lies within 0.27σ of Planck's determination 67.4 ± 0.5 km/s/Mpc, with no free parameter. The Hubble tension is now a fully structural prediction of the PDL programme.

Second, the numerical conjecture (11) for ε_G achieves 17 ppm accuracy using only structural quantities κ, $\mathcal{C} = 575/576$, and $R_{\text{tot}} = 11017$ from the proton architecture. If proved as an exact identity, it would complete the parameter-free derivation of all three fundamental couplings α, G, and k_B from the single combinatorial object $(24, 28, 930, 10087, 11017)$.

The central implication is structural. The amplitude of the Hubble tension — one of the outstanding observational puzzles of contemporary cosmology — is encoded in the integer architecture of the neutron. The neutron architecture descends by exact combinatorial derivation from the same proton quintuplet that fixes the fine-structure constant and Newton's constant. These three quantities, previously regarded as empirically independent, are projections of the same relational structure.

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